Henry Court

BS30 8YX   wr23643@bristol.ac.uk   07413 981336

henry-c2908   HCourt2908

3rd Year Computer Science MEng student at the University of Bristol with a passion for software development, with practical experience in Python, Java and C++. Seeking an internship to apply and further develop my skills in Computer Science.

Education

University of Bristol — Sept 2023 - June 2027

MEng in Computer Science

- Maintaining a First-Class average of 77% across my first two years of study, and an average of 79% in the first semester of my third year
- **Key Modules**: Machine Learning, Computer Graphics, Algorithms and Data, Programming Languages and Computation, Object Oriented Programming and Algorithms

Oldfield School, Bath — Sept 2016 - June 2023

- A Levels: A*A*AA in Maths, Further Maths, Computer Science and Physics
- GCSES: Grades 7-9 in Core Subjects, Spanish, Geography, Computer Science and Creative iMedia

Experience

Software Engineering Project Mentor — Sept 2025 - June 2026

- Held weekly meetings with a team of second year students, guiding them and giving constructive advice at all stages of the Software Engineering Project.
- Sat on the marking panel formative vivas, watching teams' presentations and asking difficult questions to prepare them for the final assessment

Research Intern — June 2025 - July 2025

University of Bristol

- Evaluated the Software Engineering Project unit via student focus groups and a school wide questionnaire
- Performed thematic analysis on data to produce a professional report, which was presented to academic staff

Software Development Intern — May 2025 - June 2025

University of Bristol

- Worked collaboratively in an independently managed student team to extend a Springboot-based customer relationship management system for the University of Bristol's Industrial Liaison Office
- Implemented strict security protocols and overhauled the user interface

Student Representative — Sept 2023 - Present

University of Bristol

- Elected Course Representative for 3 consecutive years, representing my cohort in meetings with academic staff
- Responsible for major academic structure alterations, such as a content overhaul in Computer Architecture and swapping two units to better balance workload in second year
- Organised and chaired town-hall style forums for students to bring feedback to.
- Serving as School Representative since 2024, attending faculty-level meetings to influence faculty decisions

Waiter, Bartender and Kitchen Porter — July 2023 - Present

Queen and Whippet

- Served food and drinks to guests, and maintained professional hygiene in kitchens
- Adapted to high pressure situations while working in a fast-paced team

Projects

Monster Hunter Loadout Planner — github.com/

- Built an interactive web app for planning and optimising armor loadout skills based on JSON data imported from Monster Hunter Rise
- **Technologies**: Java, Spring Boot, Thymeleaf

Cornell Box Computer Graphics Engine

- Created a graphics engine to render 3D models in 2D space, such as the cornell box model
- Included features such as soft shadows, advanced camera motions, reflective and refractive materials
- **Technology**: C++

Exploration of Machine Learning Techniques

- Explored a variety of machine learning algorithms, and evaluated their applicability to pre-existing datasets
- **Algorithms explored**: Regression Models, Neural Networks, Classifiers, Ensemble methods, K-Means and SVMs, and Hidden Markov Models
- **Technology**: Python

Optimising Lattice-Boltzmann for Isambard 3

- Optimised an implementation of the lattice-boltzmann algorithm to efficiently run on the Isambard 3 Supercomputer.
- Implemented serial and vectorised optimisations, as well as OpenMP and MPI
- **Technology**: C

VR Horror Game

- Was the VR Lead for Project: Asphyxia, a group project VR game
- Independently researched and designed a novel locomotion system to mitigate motion sickness
- **Technology**: Unity

Scotland Yard Game Logic & Character AI

- Created Java backend logic for the board game Scotland Yard
- Implemented an original AI for the Mr X character using the Dijkstra's Algorithm and Minimax
- **Technology**: Java

Distributed Game of Life

- Implemented Conway's Game of Life using a distributed system including RPC and AWS support
- **Technologies**: Go, AWS

Technologies

Languages: Java, C++, C, Python, C#, Javascript, HTML, CSS

Tools: Spring Boot, AWS, Git, Unity, Thymeleaf, Linux